New Kernel Functions can be derived From other Kernels

Sum of Kernels is a Kernel

$$K(u, v) = K_1(u, v) + K_2(u, v)$$

Scalar Product of Kernel is a kernel

$$K(u, v) = \alpha K_1(u, v)$$

Product of Kernels is a Kernel

$$K(u, v) = K_1(u, v) * K_2(u, v)$$

Combination of the above rules is a Kernel

$$K(u, v) = K_1(u, v) * K_2(u, v) + \alpha K_3(u, v)$$